

$X = \{ \overset{000}{\text{initCount}}, \overset{001}{\text{Activate}}, \overset{010}{\text{Add}}, \overset{011}{\text{Shift}}, \overset{100}{\text{End}} \}$

$I = \{ b_0, \text{EndCount} \}$

$Y = \{ \overset{000}{\text{Write [01A]}}, \overset{001}{\text{Write B}}, \overset{010}{\text{Clear P}}, \overset{011}{\text{Reset Count}}, \overset{100}{\text{Inc Count}}, \overset{101}{\text{Write P}} \}$
 $\text{Shift A } s_x, \text{ Shift B } d_x \}$
 110 11 111

X	EndCount=0	EndCount=1	EndCount=0	EndCount=1	Out put
	$b_0=0$	$b_0=0$	$b_0=1$	$b_0=1$	
000 InitCount	Activate ⁰⁰¹	Activate ⁰⁰¹	Activate ⁰⁰¹	Activate ⁰¹	Clear P, ⁰¹⁰ Write A, Write B, ResetCount, Clear P
001 Activate	Shift ⁰¹¹	Shift ⁰¹	Add ⁰¹⁰	Add ⁰¹⁰	IncCount
010 Add	Shift ⁰¹¹	Shift ⁰¹¹	Shift ⁰¹¹	Shift ⁰¹¹	add, Write P
011 Shift	Activate ⁰⁰¹	End ¹⁰⁰	Activate ⁰⁰¹	End ¹⁰⁰	Shift A s_x ¹¹⁰ Shift B d_x ¹¹¹
100 End	End ¹⁰⁰	End ¹⁰⁰	End ¹⁰⁰	End ¹⁰⁰	

X	00		01		Out put
	00	01	10	11	
000	001	001	001	001	00
001	011	011	010	010	01
010	011	011	011	011	10
011	001	100	001	100	11
100	100	100	100	100	X
101	X	X	X	X	X
110	X	X	X	X	X
111	X	X	X	X	X

$$y_1 = \bar{x}_2 x_1$$

$x_2 x_1 \backslash x_0$	0	1
00	0	0
01	1	1
11	X	X
10	X	X

$$y_0 = \bar{x}_2 x_0$$

$x_2 x_1 \backslash x_0$	0	1
00	0	1
01	0	1
11	X	X
10	X	X

$$x_2^* = \bar{x}_2 x_1 x_0 I_1 + x_2$$

$x_2 \backslash x_1 x_0$	00	01	11	10
000	0	0	0	0
001	0	0	0	0
011	0	0	1	1
010	0	0	0	0
110	x	x	x	x
111	x	x	x	x
101	x	x	x	x
100	1	1	1	1

$$x_1^* = \bar{x}_2 x_1 \bar{x}_0 + \bar{x}_2 \bar{x}_1 x_0$$

$x_2 \backslash x_1 x_0$	00	01	11	10
000	0	0	0	0
001	1	1	1	1
011	0	0	0	0
010	1	1	1	1
110	x	x	x	x
111	x	x	x	x
101	x	x	x	x
100	0	0	0	0

$$x_0^* = \overline{x_2} \overline{x_1} \overline{x_0} + \overline{x_2} x_1 \overline{x_0} + \overline{x_2} \overline{x_1} \overline{1_0} + \overline{x_2} x_1 \overline{1_1}$$

$x_2 \ x_1 \ x_0$	$1_1 1_0$	00	01	11	10
000		1	1	1	1
001		1	0	0	1
011		1	1	0	0
010		1	1	1	1
110		x	x	x	x
111		x	x	x	x
101		x	x	x	x
100		0	0	0	0

$$\begin{array}{r} 0101 + \\ 0011 \\ \hline 1000 \end{array}$$